

SYEDA ANIKA JERIN

Software Engineer

Dhaka, Bangladesh | s.anikajerin@gmail.com | +880-1634726574

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Professional Summary

Software Engineer with 6 years of experience building software systems, AI-enabled applications, enterprise platforms, and business automation solutions. Experienced in full-stack development, Odoo ERP customization, system integration, AI/ML, computer vision, and data-driven applications. Contributed to large-scale government and enterprise solutions in collaboration with national and international organizations. Research interests include computer vision, HCI, medical imaging, multimodal AI, and interactive & 3D visualization.

Technical Skills

Languages: Python, JavaScript, C++, C, SQL, Go

Software & Backend Engineering: Odoo, FastAPI, Flask, Django, REST APIs, Microservices, System Integration, ERP Customization, Business Process Automation

AI & Computer Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face Transformers, YOLOv8, YOLO-NAS, Vision Transformers, CNNs, LSTM, Deep SORT, XGBoost, LightGBM

Generative & Multimodal AI: LLMs, Multimodal Learning, RAG, Embeddings, Vector Search, LLM APIs

Data & Visualization: NumPy, Pandas, Matplotlib, Plotly, Folium, Chart.js, ECharts, amCharts, Leaflet

Frontend & 3D: JavaScript, React.js, Next.js, HTML, CSS, Three.js

Databases & Tools: PostgreSQL, MySQL, Git, GitHub, Linux, Docker, Postman, Jupyter

Professional Experience

Unisoft Systems Ltd. – Smart Technologies (BD) Ltd.

Dhaka, Bangladesh

Software Engineer

Dec 2020 – Oct 2025

- Developed production components for an **AI-driven traffic monitoring and automated fine-generation system for Bangladesh Highway Police**, part of a government project valued at approximately **BDT 72 crore**, supporting speeding and wrong-way violation detection, vehicle tracking, and automatic number plate recognition (ANPR).
- Developed and optimized computer vision pipelines using **deep learning-based object detection, multi-object tracking, and transformer-based vision models** for Bangladesh road-traffic scenarios; performed dataset analysis, augmentation, model evaluation, and integration into production inference workflows.
- Led development of core components of a **meteorological forecasting and data-archiving platform for Bangladesh Meteorological Department**, part of an approximately **BDT 30 crore** project funded by World Bank and supported by the Government of Bangladesh and the World Meteorological Organization (WMO); engineered pipelines for **SYNOP, METAR, TAF, BUFR, and SHIP** data processing, storage, analytics, and visualization.
- Designed and customized **Odoo ERP** solutions spanning Sales, Purchase, Inventory, Accounting, Manufacturing, Warehouse, HR, and operational workflows; developed custom modules and **integrated ML-based** quality monitoring and defect-analysis capabilities.
- Built **RESTful APIs, Python backend services, PostgreSQL integrations, and analytical dashboards** connecting machine learning models with enterprise applications and operational systems.
- Contributed to system architecture, requirements analysis, technical reviews, debugging, and development coordination across enterprise and AI-focused projects.

EcosoftBD – Smarter Stock

Remote

Software Engineer

Sep 2020 – Nov 2020

- Developed a machine learning pipeline for **stock-market trend analysis and time-series forecasting**, transforming raw financial data through cleaning, aggregation, exploratory analysis, and model-ready preprocessing.
- Built and evaluated **deep learning-based time-series forecasting models** on historical stock-market data to identify temporal patterns and support predictive analysis.
- Analyzed large financial datasets to identify trends and translated model outputs and exploratory findings into actionable insights for business decision-making.

Personal Projects

GSP-RenderX – Constraint-Aware Text-to-3D Generation

LLMs, Generative AI, Three.js

- Developed a **constraint-aware text-to-3D scene generation system** that transforms natural-language

descriptions into structured scene representations and interactive 3D environments.

- Built an LLM-driven **scene-graph and spatial reasoning pipeline** for object relationships and automated scene composition, including constraint enforcement, collision handling, object placement, camera fitting, and interactive visualization.

HypoTrace – Privacy-First AI Coding Coach

LLMs, Python, VS Code, SQLite

- Developed a **privacy-first AI coding coach for VS Code** that learns recurring developer mistake patterns from code-run outcomes and generates evidence-based forecasts before similar errors recur.
- Designed an **agentic learning and feedback pipeline** that generates possible explanations and counterevidence, delivers targeted coaching interventions, and evaluates subsequent runs to measure whether the intervention improved developer performance.

Cross-Modal Deep Learning for Alzheimer’s Disease Classification

3D CNN, Multimodal Learning

- Developed a multimodal deep learning framework combining **3D MRI representations with tabular clinical features** for Alzheimer’s disease classification, including preprocessing, feature fusion, model training, and evaluation pipelines.
- Explored interpretable medical imaging analysis using model visualization techniques to investigate disease-relevant regions.

Quantum-Enhanced Graph Learning for Molecular Design

GNN, Qiskit, PyTorch

- Developed a hybrid classical-quantum learning pipeline combining graph neural networks with a variational quantum prediction head for molecular property modeling, including band-gap regression and toxicity classification.

GeoMap3D – Interactive 3D Geographic Visualization Library

Python, Three.js, WebGL

- Developed a Python-oriented library for generating interactive **3D geographic visualizations** through a declarative API, enabling web-based geographic scenes without requiring users to directly implement JavaScript or GIS frontends.

Face Recognition Attendance System

Python, OpenCV, Flask, Odoo

- Built a real-time face recognition workflow integrating OpenCV and Flask with Odoo employee attendance to automate identity verification and attendance operations.

Education

Jahangirnagar University

B.Sc. in Information Technology

Savar, Dhaka, Bangladesh

March 2015 – Dec 2018

Savar Cantonment Public School & College

Higher Secondary Certificate (HSC)

Savar, Dhaka, Bangladesh

August 2014

Savar Cantonment Public School & College

Secondary School Certificate (SSC)

Savar, Dhaka, Bangladesh

March 2012

Awards & Certifications

- **Participation Award** – Ranked in the *top 31% globally in HackerRank Orchestrate (Sep 2026, Rank #924)*, a competitive AI-powered ranking for software engineers worldwide.
- **Best Impact Award** – Co-authored *COVID-Hero: Machine Learning Based COVID-19 Awareness Enhancement Mobile Game for Children*.
- **Advanced Computer Vision with TensorFlow** – DeepLearning.AI.
- **Calculus for Machine Learning and Data Science** – DeepLearning.AI.
- **Contests** – Active participant, finalist and coordinator across premier IEEE competitions: *IEEE Xtreme 24-Hour Programming*, *IEEE MadC (Mobile Application Development Contest)*, and *IEEE CS BDC Summit 2019..*
- **Government Scholarship** – Awarded for outstanding academic results in national H.S.C., S.S.C., J.S.C., and P.S.C. examinations, Bangladesh.